**ChE 451**

**"Renewable Energy Technologies"**

(Midterm Home Exam)

October 29, 2019
Due Date: November 5, 2019 (3:30 pm)

**Problem Statement:**

Your company, International Energy Consulting Group, won a contract from the UN Refugee Commission to study the technical and economics feasibilities of installing a solar PV power generating plant to supply a refugee camp that will house 20,000 refugees in North of Jordan.

Your report should include system design and sizing analysis and recommend whether the proposed solar PV system is economically favorable or not compared to conventional power generation systems (e.g. diesel electric generators) ***taking into account all direct and indirect cost as well as cost of ownership (i.e. fuel cost, maintenance and operation)***

**Data:**

1) Price of PV panels = \$0.35 /$W_p$
2) Price of batteries (Li-ion): \$250/kWhr (10 years life time)

- Make all necessary assumptions you feel necessary to complete your study
- Data on cost of diesel power generators and cost of fuel in Jordan can be assumed based on current or any similar case study with market value

**Power requirements for the Refugee camp**:

*1) Typical Refugee prefab unit (4-6 people):*

| Load Type | # Units | Rated Power (W/unit) |
|---|---|---|
| TV | 1 | 50 |
| Fan | 2 | 40 |
| LED lights | 5 | 10 |
| Small Refrigerator | 1 | 75 |

*2) Clinic*

| Load Type | # Units | Rated Power (W/unit) |
|---|---|---|
| Refrigerator | 2 | 200 |
| LED lights | 10 | 10 |
| Medical equipment | 2 | 1,500 |
| Others | 1 | 500 |

*3) Pumping water Supply*

| Load Type | # Units | Rated Power (W/unit) |
|---|---|---|
| Electric motors | 1 | 3,000 |
| Control Panel | 2 | 500 |

*4) School Lighting*

| Load Type | # Units | Rated Power (W/unit) |
|---|---|---|
| LED lights | 50 | 10 |
| Others | 1 | 500 |

*5) Street Lighting*

| Load Type | # Units | Rated Power (W/unit) |
|---|---|---|
| LED lights | 100 | 25 |

**Important Instructions:**

- You can email, mail or hand deliver exam to ChE Department office
- The exam is individual effort and no collaboration with classmates or others is permitted